

Lem transformada fourier convolucion

Lema 1. Sean $f, g \in \mathcal{L}^1(\mathbb{R})$, entonces $(f * g)^\wedge = f^\wedge \cdot g^\wedge$.

Demostración: Por la desigualdad de Young, $\|f * g\|_1 \leq \|f\|_1 \|g\|_1 < \infty$, luego

$$\forall \xi \in \mathbb{R} : (f * g)^\wedge(\xi) = \int_{\mathbb{R}} (f * g)(x) e^{-2\pi i x \xi} dx = \int_{\mathbb{R}} \left(\int_{\mathbb{R}} f(x - y) g(y) dy \right) e^{-2\pi i x \xi} dx.$$

Como ambas funciones son integrables, podemos aplicar el teorema de Fubini:

$$\begin{aligned} (f * g)^\wedge(\xi) &= \int_{\mathbb{R}} g(y) \left(\int_{\mathbb{R}} f(x - y) e^{-2\pi i x \xi} dx \right) dy \\ &= \int_{\mathbb{R}} \left(\int_{\mathbb{R}} f(x - y) e^{-2\pi i (x - y) \xi} dx \right) g(y) e^{-2\pi i y \xi} dy = f^\wedge(\xi) \cdot g^\wedge(\xi). \end{aligned}$$

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